5.2 Being optimistic under uncertainty with the Upper Confidence Bound policy

How should we account for the infinitely many possibilities for the value we may observe, when we evaluate the objective function at a particular location? Further, how should we reason about these possibilities in a simple, efficient way that is conducive to decision-making? In this section, we explore the Upper Confidence Bound (UCB) policy from MAB, which results in a BayesOpt policy of the same name.

UCB operates under the "optimism in the face of uncertainty" principle. In MAB, the idea is to use an upper bound of each slot machine's reward rate as a substitute for the true, unknown reward rate. That is, we optimistically estimate the reward rate of each machine using an upper bound of what we believe it to be, and finally choose the one with the highest upper bound.

We will first talk about this principle and how it aids our decision-making reasoning. Afterwards, we will learn how to implement the UCB policy using BoTorch and analyze its behavior.

5.2.1 Optimism under uncertainty

Let's consider a simple example to make this idea concrete. Say you wake up one day and observe that even though it's sunny outside, there are dark clouds on the horizon. You check the weather app on your phone to see if it will stay sunny throughout the day, and whether you should bring an umbrella with you to work in case it rains later. Unfortunately, the app can't tell with absolute certainty whether it will stay sunny. Instead, you only see an estimate that the probability of sunny weather is between 30 percent and 60 percent.

You think to yourself that if the probability that it will stay sunny is less than 50 percent, then you will bring an umbrella. However, you don't have a single-valued estimate here, but instead a range between 30 and 60. How, then, should you decide whether an umbrella is necessary?

A pessimist may say that since the probability of sunny weather could get as low as 30 percent, you should be on the safe side and prepare for the worst. Someone that considers the average case might look further into the app to see what the mean estimate for the probability that it will stay sunny is and make their decision accordingly. From the perspective of an optimist, on the other hand, 60 percent is enough to think that it will stay sunny, so this person wouldn't bother bringing an umbrella with them to work. These ways of thinking are shown in Figure 5.5.

In Figure 5.5, the third person's reasoning corresponds to the idea behind the UCB policy: being optimistic about the outcome of an unknown event and making decisions based on this belief. In the MAB problem, UCB would construct an upper bound of the reward rate of each slot machine and proceed to pick the machine with the highest bound.

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